# Journey of GCD

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 3 seconds |
| Memory limit: | 1024 megabytes |

In the magical kingdom of Numerian, there are $n$ cities, each with its own magical power level $a_i$. The king has built bidirectional roads between every pair of cities. For the road connecting city $u$ and city $v$, its Guardian Charm Degree (GCD) equals $\gcd(a_u, a_v)$, which is the greatest common divisor of their magical power levels.

For a traveler moving from city $x$ to $y$, the Magical Immunity Number (MIN) of the journey is the minimum GCD among all roads taken. The royal advisors consider a journey "well-immune" if and only if its MIN is maximized.

You are tasked with answering $q$ queries from the king. For each query $(x, y)$, output the maximum possible MIN among all paths from city $x$ to city $y$.

In particular, if $x = y$, the traveler never leaves the city, so the MIN is simply the city's own magical power $a_x$.

## Input

The first line contains two integers $n$ and $q$ ($1 \le n, q \le 10^6$).

The second line contains $n$ integers $a_1, a_2, \cdots, a_n$ ($1 \le a_i \le 10^6$), representing the magical power of each city.

Each of the next $q$ lines contains two integers $x$ and $y$ ($1 \le x, y \le n$), representing a query.

## Output

For each query, output a single integer representing the maximum possible MIN.

## Example

| standard input | standard output |
|---|---|
| 6 3<br>6 15 22 33 35 63<br>5 6<br>3 5<br>2 2 | 7<br>3<br>15 |

## Note

For the sample, the magical power levels are $a = [6, 15, 22, 33, 35, 63]$.

Query 1: The optimal path is $5 \to 6$, and the answer is $\gcd(35, 63) = 7$.

Query 2: The optimal path is $3 \to 4 \to 6 \to 5$. The edge values are $\gcd(22, 33) = 11$, $\gcd(33, 63) = 3$, and $\gcd(63, 35) = 7$, so the answer is $\min(11, 3, 7) = 3$.

Query 3: Since $x = y$, the traveler never leaves the city, so the answer is $a_2 = 15$.